

EMMANUEL CHUKWUEBUKA MOGHALU

Data Engineer | Backend Systems Engineer

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PROFESSIONAL SUMMARY

Data and backend engineer with 4+ years of experience across full-time, part-time, contract, and freelance roles, building production-grade APIs, web applications, ETL/ELT pipelines, and analytics platforms. Deep working knowledge of the Nigerian and broader African fintech ecosystem, including multi-PSP integration (Paystack, Flutterwave, Moniepoint, OPay, Interswitch), CBN regulatory requirements, and NDPR compliance. Comfortable owning a system end-to-end — from PRD and data architecture through implementation, testing, and dashboarding — and communicating technical tradeoffs clearly to cross-functional stakeholders.

CORE TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Backend Engineering: Django, FastAPI, REST APIs, Microservices, API Design, Authentication

Data Engineering: dbt, Dagster, Apache Spark, PostgreSQL, ETL/ELT Pipelines, Data Modeling, Data Warehousing, pgvector, Meilisearch

Analytics & BI: Pandas, Streamlit, Grafana, Power BI, Plotly, KPI Development, Dashboarding

DevOps & Infrastructure: Docker, Git, GitHub Actions, CI/CD, Linux, Monitoring & Logging

Databases: PostgreSQL, SQL Optimization, Query Performance Tuning

Domain Expertise: Nigerian/African Fintech (CBN & NDPR compliance, PSP integration), System Design, Distributed Systems

PROFESSIONAL EXPERIENCE

Software & Data Engineer (Contract)

Jan 2026 – Feb 2026

MedVax Health

- Developed backend services supporting secure healthcare workflows, authentication, payment processing, and transactional systems.
- Built reliable data processing pipelines to improve structured data flow across product features.
- Investigated production issues using structured debugging, logging, and root cause analysis techniques.
- Improved backend stability through API optimization, monitoring, and proactive issue resolution.

Technologies: Python, FastAPI, PostgreSQL, REST APIs, Docker

Data Operations & Systems Engineer

Mar 2024 – Oct 2024

SETRACO Nigeria Ltd (completed alongside final year of B.Eng.)

- Designed and implemented centralized reporting solutions that replaced manual Excel-based workflows, reducing reporting effort by approximately 40%.
- Built SQL-powered dashboards that improved operational visibility and accelerated management decision-making.
- Maintained 99.9% data accuracy across production reporting systems through data validation and quality assurance.
- Diagnosed and resolved SQL pipeline failures and data inconsistencies, improving overall operational reliability.
- Optimized SQL queries and backend processes to improve reporting performance and scalability.

Technologies: Python, SQL, React (Next.js), PostgreSQL, Dashboard Development, Data Analysis

Freelance Software & Data Engineer

Oct 2020 – Mar 2024

Self-Employed (concurrent with B.Eng., Caritas University)

- Delivered web development and data solutions for independent and small-business clients, managing full project lifecycles from proposal through delivery.
- Authored client-facing technical proposals and scoped project requirements.
- Designed and shipped a personal portfolio site (React, TypeScript, Tailwind CSS, Framer Motion).
- Sustained continuous hands-on engineering work throughout a full-time engineering degree program.

Technologies: Python, JavaScript/TypeScript, React, SQL

Full-Stack Software Engineer (Part-Time)

Aug 2019 – Aug 2020

TAC AFRICA (concurrent with AfriHUB ICT Solutions, below)

- Developed and maintained web applications serving over 50,000 users, contributing to improved platform stability and user experience.
- Optimized backend APIs and SQL queries, reducing response latency by approximately 45%.
- Investigated and resolved production issues through structured debugging, log analysis, and database optimization.
- Collaborated with designers and product teams to implement new features and improve application performance.

Technologies: Python, JavaScript, SQL, REST APIs, PostgreSQL

AfriHUB ICT Solutions (undertaken alongside the National Innovative Diploma program, below)

- Designed and implemented software solutions for academic and business clients using modern web technologies.
- Diagnosed and resolved application, database, and integration issues in production environments.
- Supported deployment, system integration, and maintenance of enterprise applications.
- Assisted in building secure database structures and backend business logic for client projects.

Technologies: Python, SQL, HTML, CSS, JavaScript, MySQL

SELECTED PROJECTS — BUILT & DEPLOYED

Cross-Border Mobile Money Reconciliation Engine

- Designed a reconciliation platform spanning eight Nigerian and African payment service providers (Paystack, Flutterwave, M-Pesa, Moniepoint, OPay, Interswitch, Remita, Paga), addressing PSP webhook unreliability and CBN/NDPR compliance requirements.
- Authored a full pre-engineering documentation suite: PRD, Data Architecture Blueprint, ERD, Data Dictionary, Technical Design Document, API Specification, Data Governance Policy, and QA Strategy.
- Built and shipped a React dashboard and landing page for the platform.

ULTRA-NEWS V3 — News Aggregation Platform

- Rebuilt a legacy Django news-aggregation monolith around an ingestion pipeline using pgvector embeddings and Meilisearch full-text search.
- Diagnosed a structural clustering failure — exact-title matching could not group synonymous financial terms (e.g., “CBN” vs. “Central Bank of Nigeria”) — and redesigned the system around a swappable ClusterScorer protocol (exact-match → token-overlap → embedding-based).
- Seeded and verified live RSS ingestion from seven African financial news sources.
- Shipped a Phase 1 implementation alongside a full UI/UX design system.

Global Rate Limiter as a Service — Vega IT Competition

- Built a high-availability, cluster-safe rate limiting service for third-party API calls as a competition qualification submission.
- Implemented per-client and tier-based (free/premium) rate limits with fail-safe behavior on database/cache outage.
- Delivered asynchronous logging for billing and analytics, plus a usage dashboard.
- Shipped with unit tests (including race-condition and load tests), a full Docker Compose setup, an architecture diagram, and a README.

Modern Enterprise Data Warehouse

- Designed and implemented an analytics platform processing over 1.5 million records using a Medallion (Bronze–Silver–Gold) architecture.
- Built scalable ETL/ELT pipelines using dbt and Dagster, with 21 automated data quality tests.
- Developed analytics-ready data models and dashboards supporting executive reporting and KPI tracking.

ARCHITECTURE & SYSTEMS DESIGN

Design-stage work; not yet running in production.

CBN Data Residency Migration Reference Architecture

- Designed a reference architecture for migrating Nigerian payment transaction data to local storage ahead of the CBN’s January 2027 compliance deadline (Circular PSS/DIR/PUB/CIR/001/004).

Distributed Smart Meter & Inverter Telemetry Platform

- Produced blueprint-stage architecture for a real-time IoT telemetry platform using Flink, Iceberg, and TimescaleDB.

EDUCATION

Bachelor of Engineering (B.Eng.), Computer Engineering

2021 – 2025

Caritas University, Enugu State, Nigeria — CGPA: 4.10 / 5.00

National Innovative Diploma (NID), Software Engineering

2018 – 2020

AfriHUB ICT Solutions, Abuja, Nigeria — Graduated with distinction, Top 5% of class

PROFESSIONAL CERTIFICATIONS

DataCamp Data Engineer Certification — Completed comprehensive hands-on training in modern AI & data engineering, May 2026.